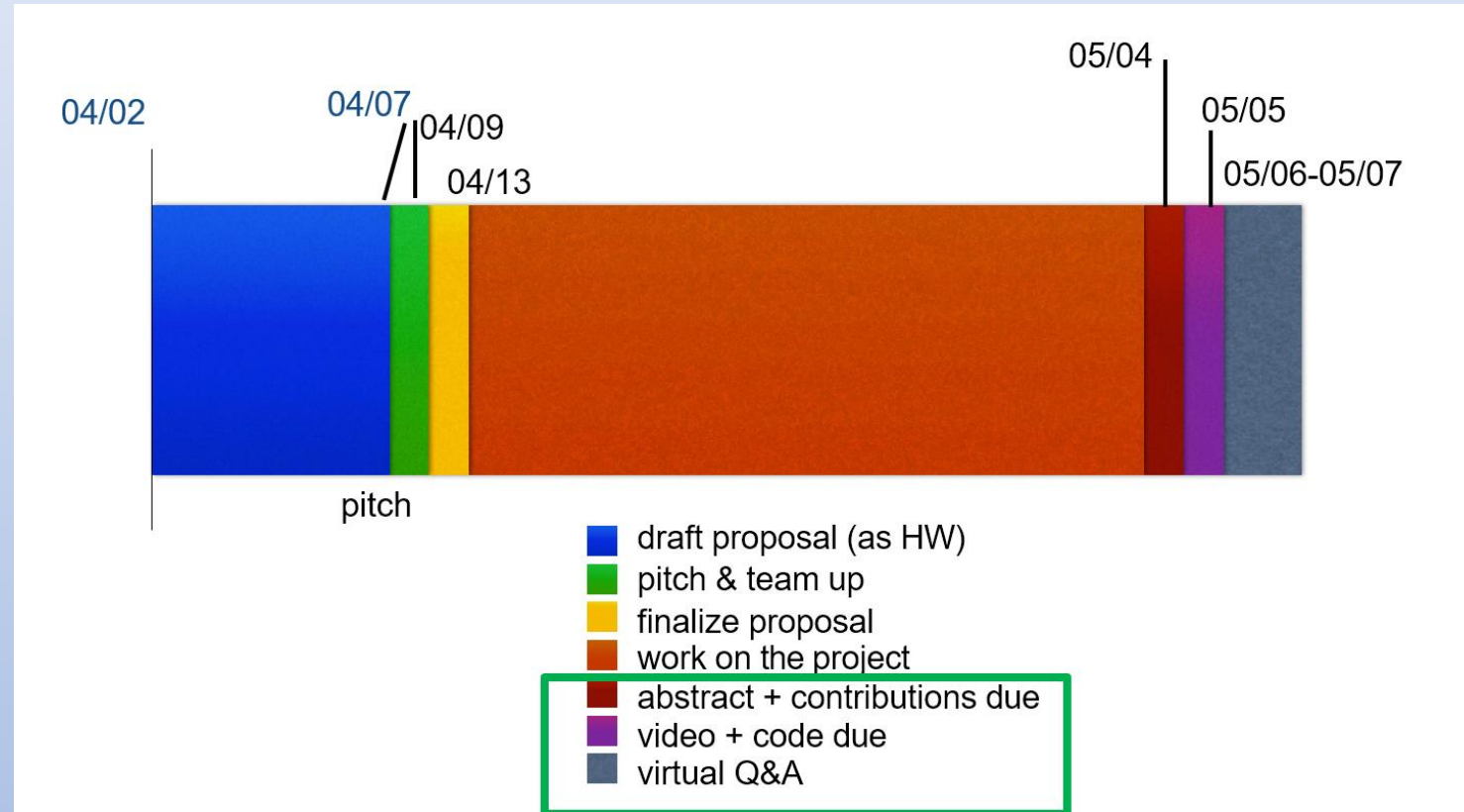
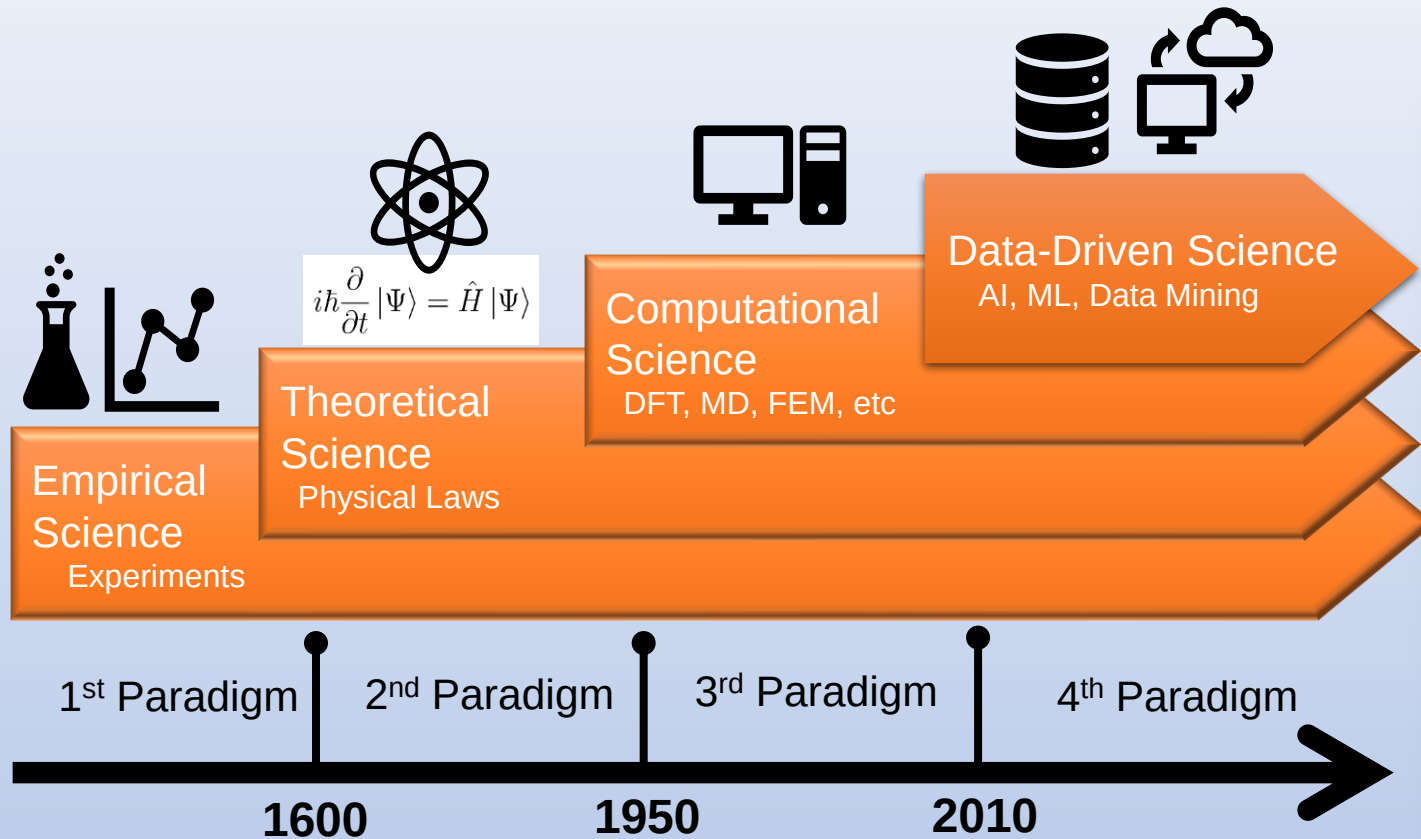


Announcements

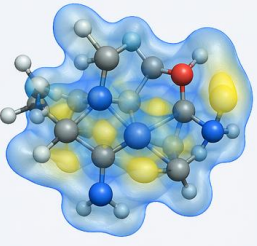
- Midterm 2 Scores will be posted before next week
- HW 10 (last HW) due on April 30th- **Capacitor in a box**
- Final Exam deadlines-



The Evolution of Scientific Discovery- Role of Computational Methods in Physics



DFT
 Density Functional Theory

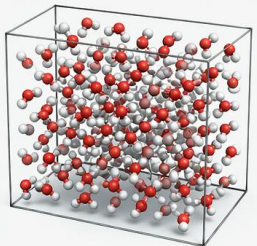


$$E[\rho] = T_s[\rho] + E_{ext}[\rho] + E_H[\rho] + E_{xc}[\rho]$$

- Quantum mechanical method
- Electrons treated explicitly
- Ground-state properties
- Based on electron density $\rho(r)$

Applications:
 Electronic structure, bonding, catalysis, materials design

MD
 Molecular Dynamics

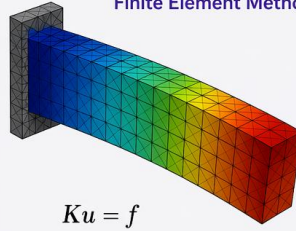


$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = -\nabla_i V(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$$

- Atoms/molecules classical
- Newton's equations of motion
- Time evolution of systems
- Interatomic potentials

Applications:
 Liquids, biomolecules, polymers, transport, thermodynamics

FEM
 Finite Element Method

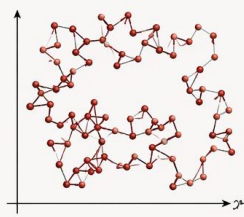


$$Ku = f$$

- Continuum mechanics
- Geometry discretized into elements
- Solves PDEs (e.g., elasticity, heat transfer)
- Highly versatile

Applications:
 Structures, vibrations, fluids, heat transfer, multiphysics

MC
 Monte Carlo Method



$$\langle A \rangle \approx \frac{1}{N} \sum_{i=1}^N A(x_i)$$

- Statistical sampling
- Random numbers
- Evaluates averages
- Useful for high-dimensional or complex systems

Applications:
 Statistical physics, finance, radiation transport, optimization

- Computational Sciences- DFT, MD, MC, FEM lead to dramatic advances in science and technology

Example-Solving the Schrödinger Equation

$$E\psi = H\psi$$

where

$$H = \underbrace{-\sum_i \frac{\hbar^2}{2m_e} \nabla_i^2}_{\text{KE of electrons}} - \underbrace{\sum_k \frac{\hbar^2}{2m_k} \nabla_k^2}_{\text{KE of nuclei}} - \underbrace{\sum_i \sum_k \frac{e^2 Z_k}{r_{ik}}}_{\text{Coulombic attraction between nuclei and electrons}} + \underbrace{\sum_i \sum_j \frac{e^2}{r_{ij}}}_{\text{Coulombic repulsion between electrons}} + \underbrace{\sum_k \sum_l \frac{Z_k Z_l e^2}{r_{kl}}}_{\text{Coulombic repulsion between nuclei}}$$

Density Functional Theory & Algorithm

The Kohn-Sham Equations

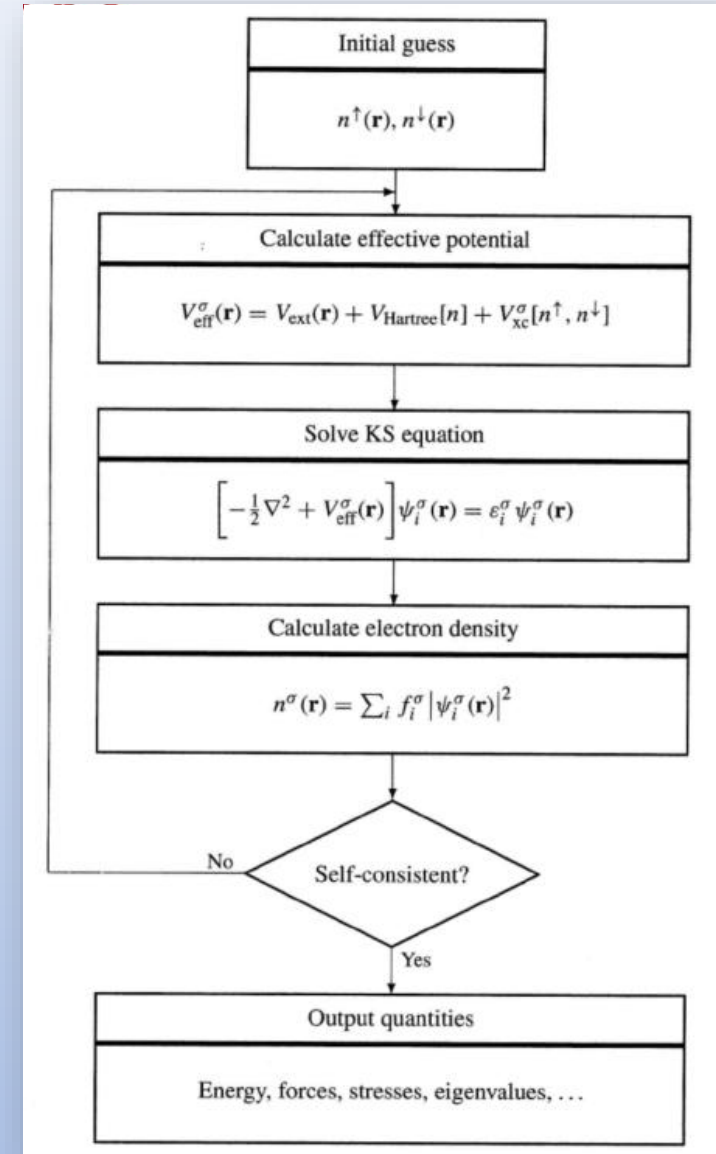
$$E[\rho(\mathbf{r})] = \sum_i \left[-\frac{1}{2} \int \psi_i^*(\mathbf{r}) \nabla^2 \psi_i(\mathbf{r}) d\mathbf{r} - \sum_k \int \psi_i^*(\mathbf{r}) \frac{Z_k}{r_{ik}} \psi_i(\mathbf{r}) d\mathbf{r} \right] \\ + \sum_i \frac{1}{2} \int \int \frac{\rho(\mathbf{r}')}{|\mathbf{r}_i - \mathbf{r}'|} |\psi_i(\mathbf{r})|^2 d\mathbf{r}' d\mathbf{r} \\ + E_{xc}[\rho(\mathbf{r})]$$

KS one-electron operator

$$h_i^{KS} = -\frac{1}{2} \nabla^2 - \sum_k \frac{Z_k}{r_{ik}} + \frac{1}{2} \int \int \frac{\rho(\mathbf{r}')}{|\mathbf{r}_i - \mathbf{r}'|} d\mathbf{r}' + V_{xc}[\rho(\mathbf{r})]$$

$$h_i^{KS} \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r})$$

$$\rho(\mathbf{r}) = \sum_i |\psi_i(\mathbf{r})|^2$$



DFT Compute Intensive

The Kohn-Sham Equations

$$E[\rho(\mathbf{r})] = \sum_i \left[-\frac{1}{2} \int \psi_i^*(\mathbf{r}) \nabla^2 \psi_i(\mathbf{r}) d\mathbf{r} - \sum_k \int \psi_i^*(\mathbf{r}) \frac{Z_k}{r_{ik}} \psi_i(\mathbf{r}) d\mathbf{r} \right] \\ + \sum_i \frac{1}{2} \iint \frac{\rho(\mathbf{r}')}{|\mathbf{r}_i - \mathbf{r}'|} |\psi_i(\mathbf{r})|^2 d\mathbf{r}' d\mathbf{r} \\ + E_{xc}[\rho(\mathbf{r})]$$

KS one-electron operator

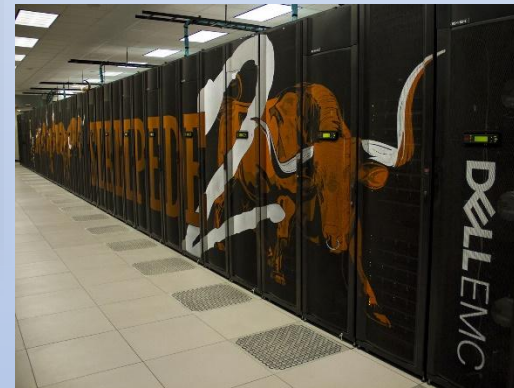
$$h_i^{KS} = -\frac{1}{2} \nabla^2 - \sum_k \frac{Z_k}{r_{ik}} + \frac{1}{2} \iint \frac{\rho(\mathbf{r}')}{|\mathbf{r}_i - \mathbf{r}'|} d\mathbf{r}' + V_{xc}[\rho(\mathbf{r})]$$

$$h_i^{KS} \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r})$$

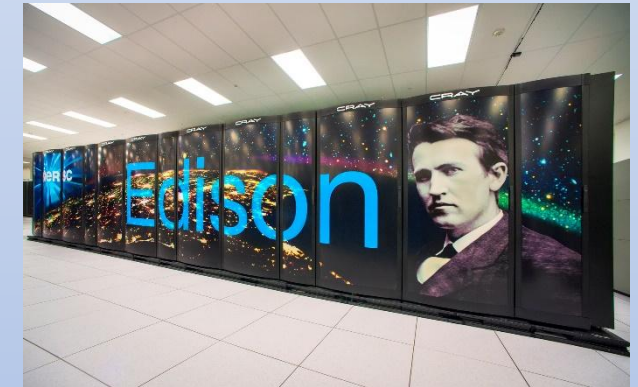
$$\rho(\mathbf{r}) = \sum_i |\psi_i(\mathbf{r})|^2$$

- In principle exact, the form of exchange correlation is unknown
- A lot of numeric related to basis sets, treatment of core/valence electrons, exchange-correlation, solving differential equations, finding eigen values etc.
- Actual production DFT codes contain tens to hundreds of thousands of lines of code

Computational resource intensive

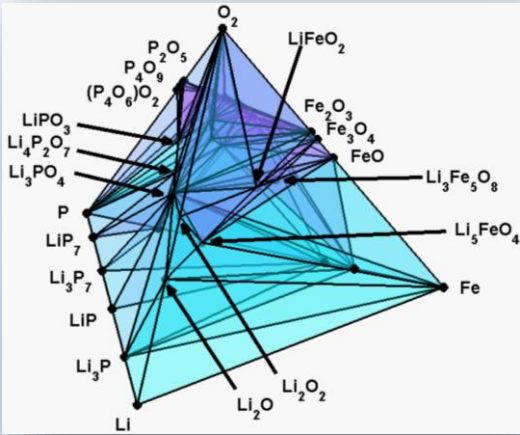


4,200 KNL nodes, 285,600 cores

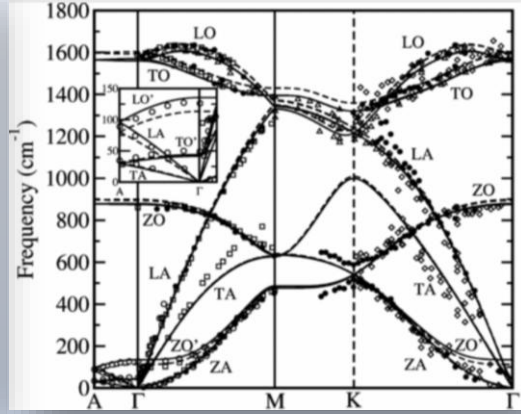


5,200 Cray XC30 nodes, 134,064 cores

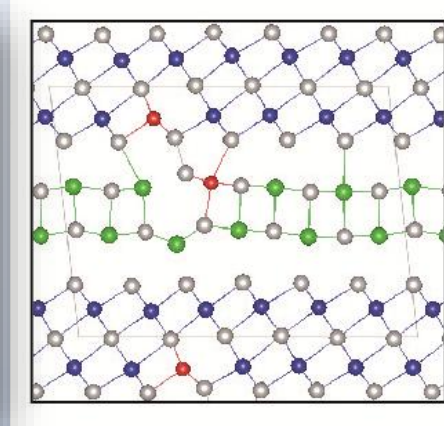
Properties Computed From DFT



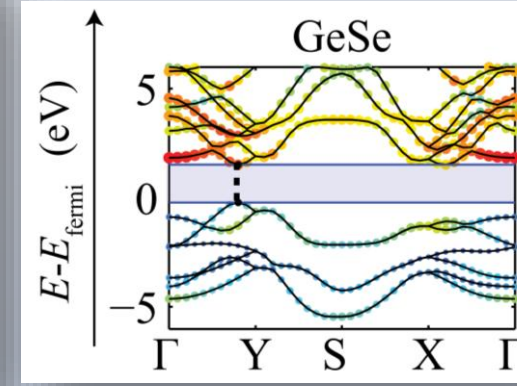
Chemistry of Materials, 2008, 1798



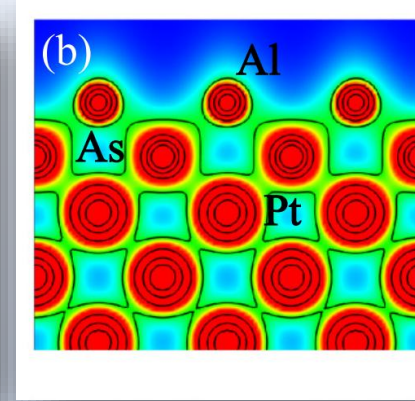
Phys. Rev. B **71**, 205214, 2005



Ren, Singh et al., Nano Lett.
2016, 16



Appl. Phys. Lett. 105, 042103
(2014)



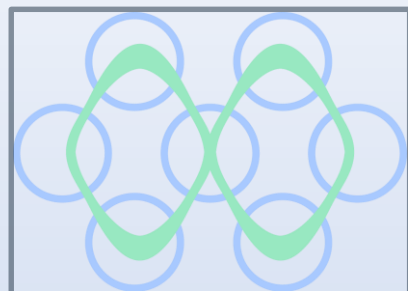
Singh et. al PRB **89**, 245431
(2014)

Computed by DFT

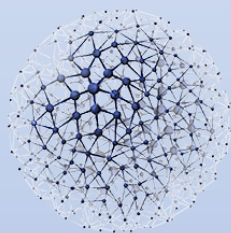
- Ground state energy
- Adsorption, chemical reactions
- Elastic constants
- Phonon Spectra
- Transition energy barriers
- Defects- vacancies, interstitials, substitution
- Band structure, band alignment, effective mass of electron and holes, work function
- Solvation energies
- Dielectric constants
- Absorption spectra, Raman spectra

Transferable to
large-scale
simulations

DFT Databases



Materials Project



AFLow

Automatic - FLOW for Materials Discovery



>83,623

Inorganic
Compounds

>53,340

Bandstructures

>21,954

Molecules

>530,243

Nanoporous
Materials

>7,122

Elastic Tensors

>1,077

Piezoelectric
Tensors

Director

Kristin Persson

Assistant Professor, Department of Materials Science and Engineering,
University of California at Berkeley

Staff Scientist, Lawrence Berkeley National Laboratory

Co-Investigators

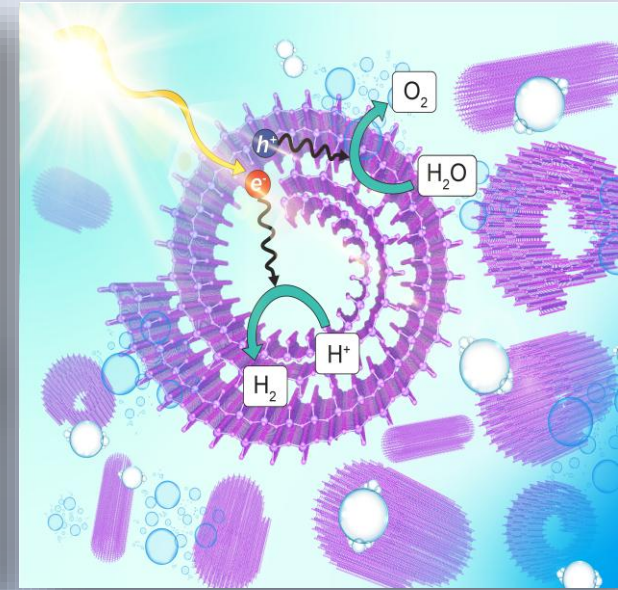
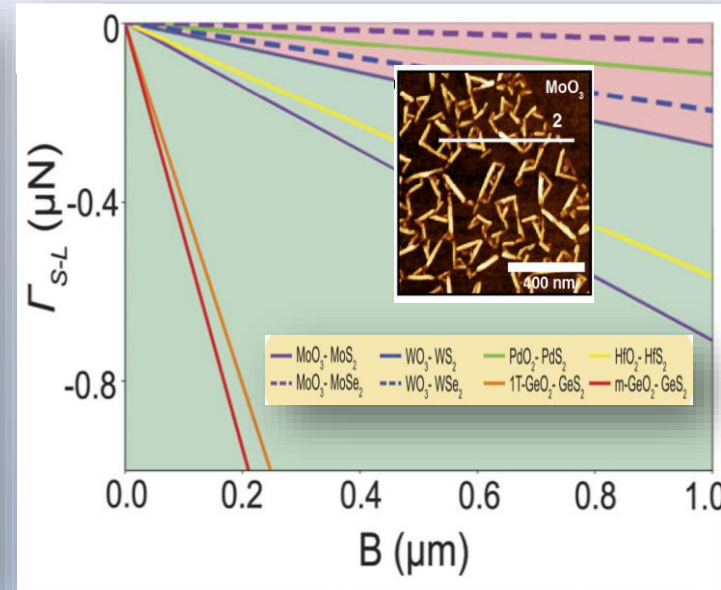
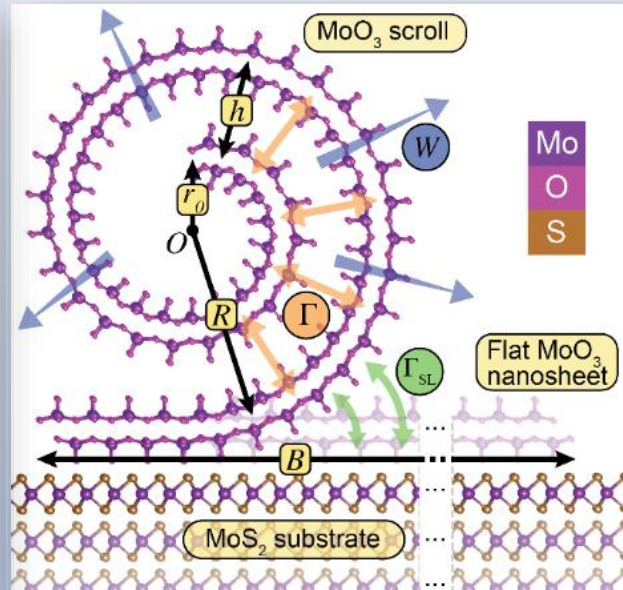
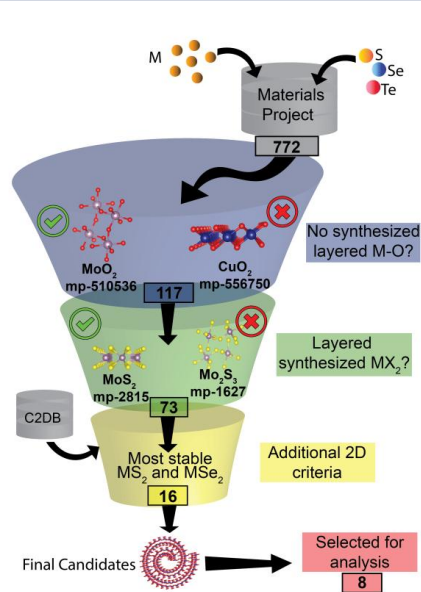
Mark Asta	Faculty Scientist, LBNL
Gerbrand Ceder	Senior Faculty Scientist, LBNL
Daryl Chrzan	Faculty Scientist, LBNL
Daniel Gunter	Staff Scientist, LBNL
Geoffroy Hautier	Asst Professor, UCL Belgium
Gian-Marco Rignanese	Professor, UCL Belgium
Anubhav Jain	Staff Scientist, LBNL
Peter Khalifah	Faculty Scientist, BNL
Lane Martin	Faculty Scientist, LBNL
Jeffrey Neaton	Senior Faculty Scientist, LBNL
Shyue Ping Ong	Asst Professor, UCSD

In Collaboration With:

GitHub users with pull requests merged to MP code repositories.

ATNDIaye	dongsenfo	Paulin
Xin Chen	David Dotson	petousis
Michael Lamparski	Donny Winston	Janakiraman Balachar
Gabriel Antonius	antonio	renpj
Holland	flub	Richard Gowers
Hannan Tang	francoi	Richard Tran
JPalakapilly	Germain Salvato Valverde	Saurabh Bajaj
Liam Deacon	Gilberto Pastorello	sbktw
Rami Lehe	gmatteo	Stephen Dacek
Rooooommmmmllll	Guido Petretto	Shaun Ziqin Rong
Tinaatucsd	Geoffroy Hautier	shenj389
Zhuoying	Henrik Rusche	shredd
Zhi Deng	jakirkham	Shyam Dwaraknath
Alan Dozier	John Dagdelen	Shyue Ping Ong
Alreza Faghanina	lek-Heng Chu	smithmacksie90
Viet-Anh HA	kathermetatimer2013	KaLiu
Alex Dunn	Ketan Bhatt	Liz Decolvenaere
Arunima Singh	Koki MURAOKA	Tonnam Balankua
Murathan Aykol	Liam Damewood	Patrick Huck
Ben Afferbach	leioheng	ucsdmg
bfoster-ibf	lordzappo	Alex Ganose
Benjamin Morgan	Marco Esters	Shishin Mo
Tess	Kiran Mathew	vonverko
Brandon Bocklund	Bharat Medasani	waffle-iron
Chi Chen	moodawc	Will Richards
Wei Chen	Andrius Merkys	Brandon Wood
ongmtam	Miao Liu	William Scullin
Anubhav Jain	Miguel Dias Costa	Xiaohui Qu
czhengsci	Michael Humbert	yihaxiao02
Dan Gunter	Matthew Horton	zacharygibbs
davidwarquiers	Joseph Montoya	Zhenbin Wang
Danny Broberg	Nicholas Wagner	Zachary Ulissi
David Cossey	ndardenne	
Danil Kitchaev	Nils Zimmermann	

Data-Driven Approach to Discovery of Oxide Nanoscrolls for Water-Splitting



Data Mining for
Precursors

DFT-based
Physio-
Mechanical
Models

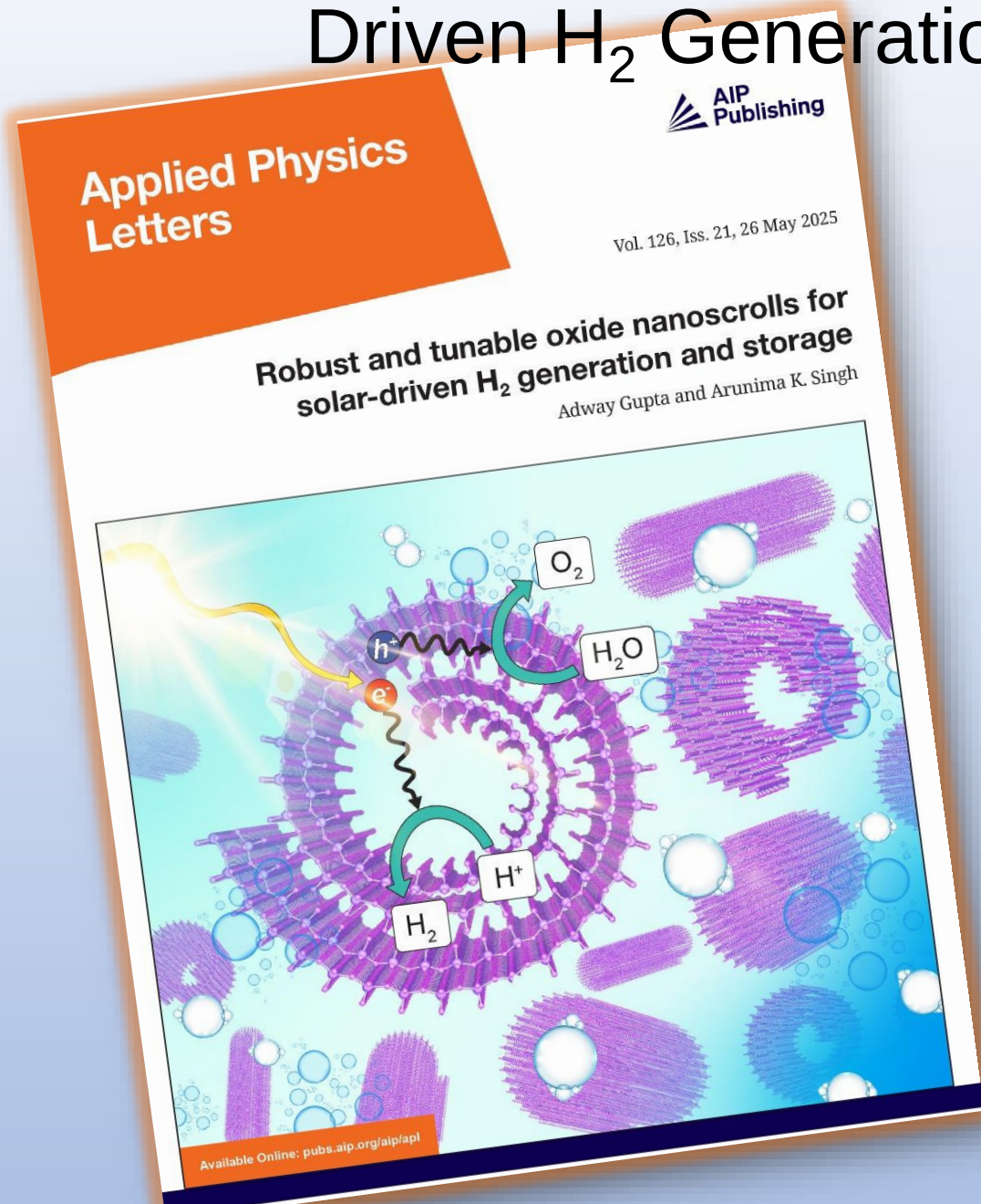
Prediction and
Experimental
Validation

Demonstrated
Applications-DFT,
AIMD, GW-BSE

A. Gupta, A. K. Singh, *APL Materials* 13, 011123 (2025)

A. Gupta, A. K. Singh, *Applied Physics Letters*, 125, 213901 (2025)

Robust and Tunable Metal Oxide Nanoscrolls for Solar-Driven H_2 Generation and Storage



U.S. DEPARTMENT OF
ENERGY

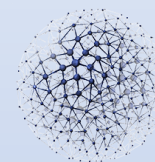
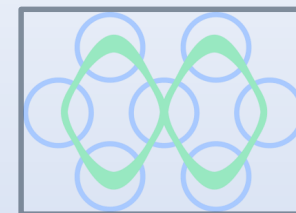
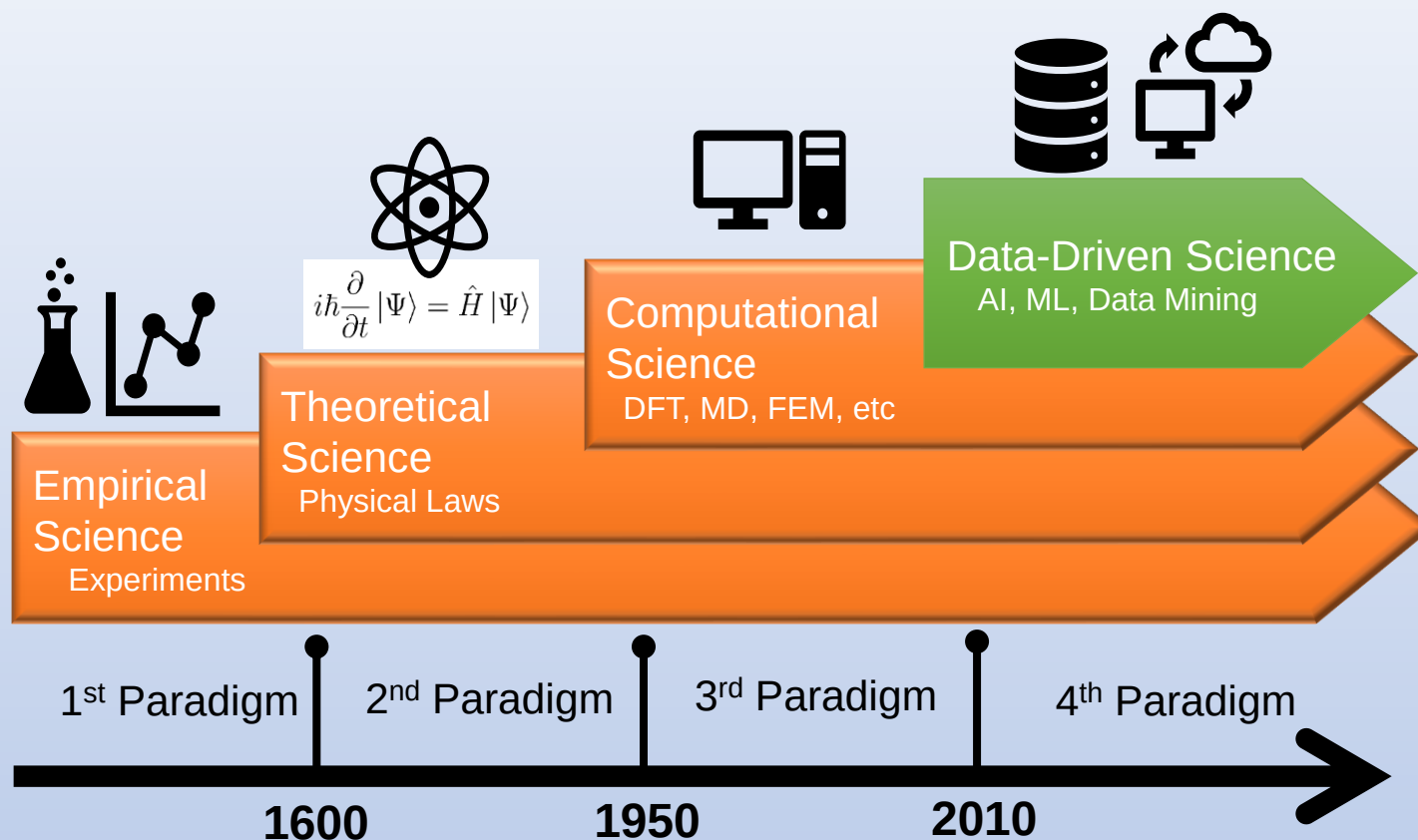


ACCESS

NERSC

ASU
Arizona State
University

AI-Enabled Science and Engineering

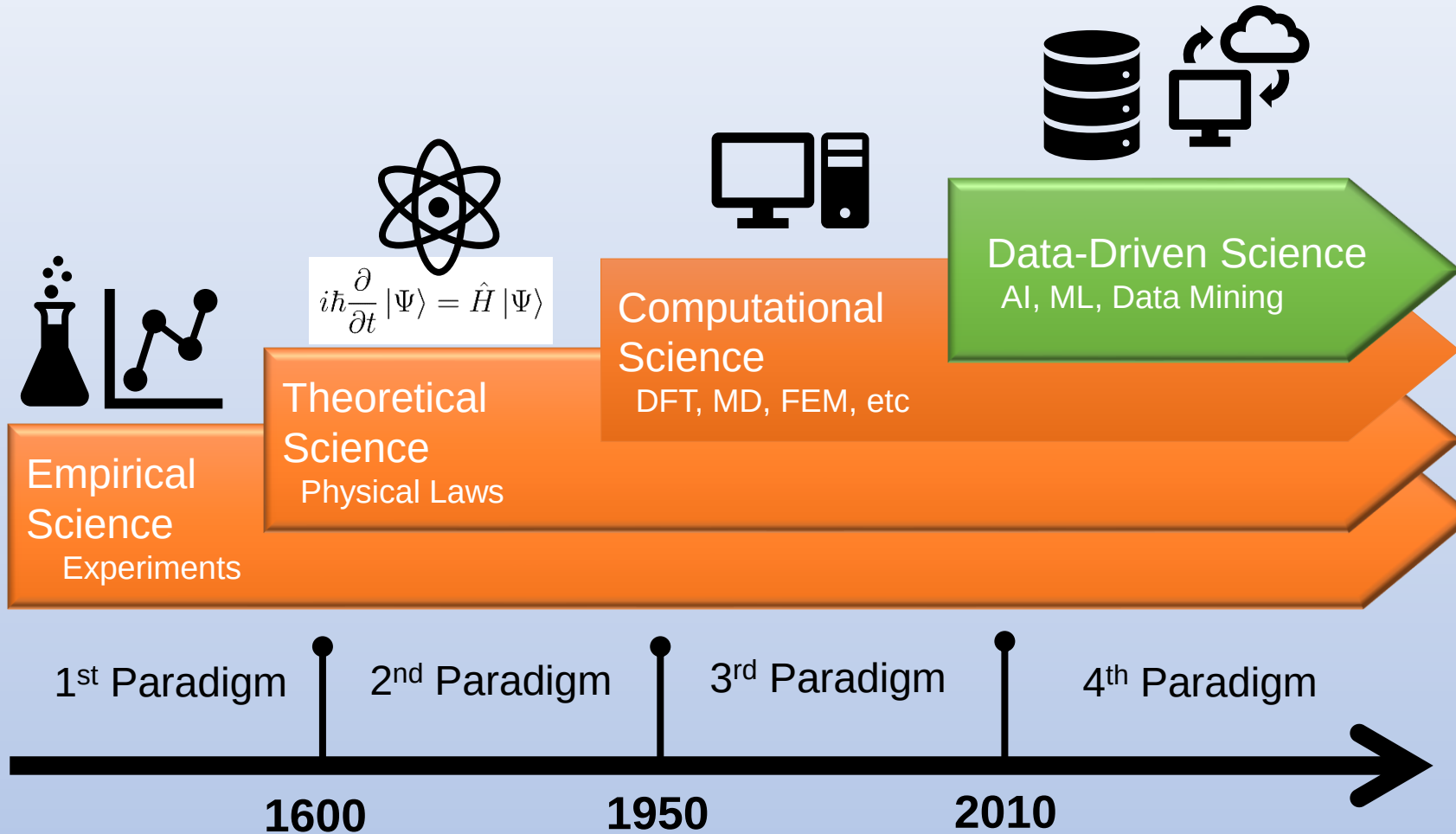


AFLOW
Automatic - FLOW for Materials Discovery



- Materials Genome Initiative (MGI) started in 2011
 - **goals** of accelerating the discovery, manufacturing, and deployment of materials **twice as fast and at a fraction of the cost**
- Led to several first-principles computation-based materials databases which curate structure and properties of thousands of **bulk materials**

AI-Enabled Science and Engineering- Tool Kit



- Existing database used for science and engineering
- Software Infrastructure
- Data Curation and Dissemination
- Physics-guided machine learning models

Machine Learning in Materials: A Paradigm Shift

- Traditional workflow: DFT → MD → FEM → Experiment
- Key bottlenecks: limited scales, huge design space, expensive experiments
- ML enables fast surrogate models and large-scale exploration
- Shift from physics-only → physics + data-driven discovery
- ML Methods in Materials Science
 - Surrogate models (GP, GNNs)
 - ML interatomic potentials
 - Active learning / Bayesian optimization
 - Physics-informed ML

Current Impact: Closed-Loop Materials Discovery

- Accelerated discovery via ML screening
- Inverse design: target $\rightarrow$ material
- Active learning with experiment feedback
- Self-driving lab paradigm emerging

Future: Agentic, Physics-Aware Materials Design

- Self-driving labs and autonomous optimization
- Closed-loop theory $\leftrightarrow$ experiment $\leftrightarrow$ device
- Multi-modal data fusion
- Agentic AI orchestration